

SPOORTHY KUMBASHI RAGHAVENDRA

Texas A&M University, College Station

spoorthy.kr@tamu.edu | +1-(979)-599-4184 | [linkedin.com/in/spoorthy-k-r/](https://www.linkedin.com/in/spoorthy-k-r/) | github.com/Spoorthy-K-R

ACADEMIC QUALIFICATION

Texas A&M University

Master's Degree in Computer Science

Courses: Deep Learning, Natural Language Processing, Analysis of Algorithms

Aug 2025 - May 2027

CGPA: 4.0/4.0

PES University

B.Tech in Electronics and Communication Engineering – **Gold Medalist**

Courses: Data Structures, Machine Learning, Artificial Intelligence, Wireless Networks and Communication

Aug 2019 - May 2023

CGPA: 9.68/10

TECHNICAL SKILLS

Languages, Tools & Frameworks: Python, Java, C, SQL, React, HTML, CSS, MATLAB, TensorFlow, Keras, Flask, Django, Spring Boot, Cucumber, JUnit, Maven, IntelliJ, Jenkins, Git, Docker, Kubernetes

Coursework and Interests: Deep Learning, Natural Language Processing, Data Structures and Algorithms, Artificial Intelligence, Image Processing, Software Engineering, Data Wrangling & Exploratory Analysis

WORK EXPERIENCE

Software Engineer I, JP Morgan Chase & Co., Bengaluru

June 2023 – July 2025

- Designed and deployed scalable **Spring Boot** microservices to replace a 40-year-old mainframe for processing 250–300k daily volumes of CLS Foreign Exchange trades worth \$10–15B, reducing operational risk and maintenance costs while improving system resilience.
- Developed **Java libraries** for multi-**MQ** consumers and **Kafka** Ack/Nack handling in a microservices architecture, reducing message trace debugging time by 4–5 hours per service and improving observability across the platform.
- Migrated large-scale Oracle databases to secure **cloud infrastructure**, ensuring compliance with strict data-security standards with minimal downtime and optimized storage and query performance through **defragmentation** tasks.
- Developed **Python-automation tools** for large-scale data ingestion, validation, and performance monitoring, saving the QA team 7–8 hours weekly.
- Resolved vulnerabilities and code smells identified by **SonarQube** and **Fortify** across the services increasing the security score from 65% to 80+% and included this as a necessary step in the CI/CD pipeline.
- Converted the **REST APIs** between the UI and microservice backend to **GraphQL** reducing frontend data processing complexity and improving load time by ~100ms.

Software Engineering Intern, JP Morgan Chase & Co., Bengaluru

Feb 2023 – May 2023

- Led end-to-end development** of a real-time monitoring platform for Kubernetes-based microservices, building a responsive React frontend and a scalable Spring Boot backend.
- Implemented dynamic pod scaling, live log streaming and service health monitoring to enable granular tracking of trades and real-time failure alerts, cutting resolution time by 40% and enhancing support team efficiency.

Machine Learning Project Trainee, BGSW, Bengaluru

June 2022 – July 2022

- Engineered a **Natural Language Processing**-driven preprocessing pipeline using spaCy, NLTK and regex to perform OCR-like parsing of symbolic reports into structured data by extracting key test parameters, enabling downstream ML model training.
- Developed a custom **position-based vectorizer** to encode term locations relative to a pivot element, boosting parsing accuracy of equations into LHS/RHS components by ~30% and ensuring accurate execution and validation of tests.
- Trained and optimised **Random Forest** and **SVM** models on structured Electric Vehicle component testing dataset achieving 85%+ accuracy in auto-generating test instructions, reducing manual effort by 60%.

PROJECTS

Deep Learning for Signal Detection in RSMA Receiver

Lead Author @ IEEE NCC 2023

- Developed an **LSTM**-based signal detection model for beyond **5G wireless networks** that uses Rate-splitting Multiple Access (RSMA).
- Used temporal sequence modeling to reduce bit error rates and outperform conventional signal detection algorithms in noisy, multi-user beyond-5G channels.

Deep Learning for Radiogenomic classification of Brain Tumor

Lead Author @ IEEE INDICON

- Led end-to-end research on transfer-learning with **ResNet** and **AlexNet** Deep Learning models to predict status of genetic biomarker from MRI scans for prediction of chemotherapy response in brain tumor patients.
- Optimised ResNet performance via 3D MRI slice extraction and advanced image augmentation/processing techniques, achieving **82% classification accuracy** on RSNA-MICCAI dataset for biomarker status prediction from MRI scans.

LLM-Powered Financial Filings & Stock Analysis

- Built an LLM-powered **Retrieval-Augmented Generation (RAG)** pipeline that downloaded and parsed SEC 10-K filings of publicly traded companies, extracted key financial metrics and indexed them in a vectorized knowledge base.
- Integrated retrieved filing insights with **stock market data** to generate multi-year **financial insights**, qualitative valuations and comparative portfolio analysis for investment decision-making.

Comprehensive Topic Summarizer using NLP

April 2022 – June 2022

- Built a **topic summarization platform** that scraped raw web content using Python and BeautifulSoup, extracting relevant text, images, and references from multiple sources.
- Integrated a **BART transformer model** for abstractive summarization, generating coherent multi-source topic overviews, and deployed the system with a Flask front-end for interactive use overviews.

AWARDS AND ACHIEVEMENT

Extracurricular:

- **Teaching Assistant:** Organized lab sessions, graded assignments and mentored 40+ students for Digital Image Processing course, leading seminars on applications of deep learning for image classification (**Aug 2022-Dec 2022**)
- **Class Representative:** Voiced student concerns to the department chair and effectively communicated departmental decisions to classmates (**Aug 2020 - May 2023**)

Awards:

- **Gold Medalist**, awarded for achieving the highest academic performance in the department.
- **Prof CNR Rao Merit scholarship**, awarded four times for ranking in the top 5% of the department
- **Dr MRD Merit Scholarship**, awarded twice for ranking in the top 20% of the department

Publications:

- **K. R. Spoorthy**, S. B. Hudalimath, S. Medatati, V. Sinchana, A. K. Kowshik, and S. Gurugopinath. "A Two-Tier Deep Neural Network Detector for Two-User Rate-Splitting Multiple Access Systems." *2023 National Conference on Communications (NCC)*, Guwahati, India, 2023. (DOI 10068068)
- **K. R. Spoorthy**, Akash R. Mahdev, Vaishnav B, and Shruthi M. L. J. "Deep Learning Approach for Radiogenomic Classification of Brain Tumor." *2022 IEEE 19th India Council International Conference (INDICON)*, Kochi, India, 2022. (DOI 10039760)

Certifications:

- Machine Learning Onramp (MathWorks, Jan 2022)
- Python for Data Science and Machine Learning (Udemy, Jun 2021)
- Machine Learning (Stanford Online, May 2021)